Automated Essay Scoring based on Multi-layer Encoders

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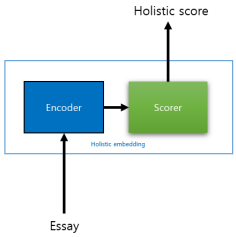
Abstract

To improve existing essay scoring models using a holistic embedding or trait embeddings, we propose a novel prompt-specific holistic and trait scoring model using multi-layer encoders that generate different types of embeddings, including trait, homogeneous, and heterogeneous embeddings. Using these diverse embeddings, the proposed model captures the contextual structure of an essay at multiple levels, enabling a more accurate prediction of essay scores. Compared to existing models that rely solely on holistic or trait embeddings, the experimental results show that the proposed model outperforms state-of-the-art models, improving overall QWK by approximately 6.3% on the ASAP/ASAP++ dataset and by approximately 15% on the AIHUB dataset.

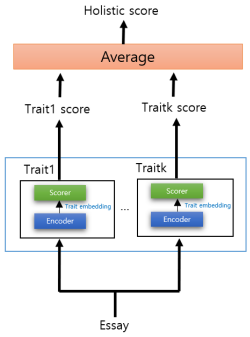
1 Introduction

Traditional Automated Essay Scoring (AES) is a task that evaluates essays based on how well they align with a given prompt, and with the advent of transformer- and large language model (LLM)-based AES models, it has achieved significant progress. Most existing models are categorized into one of the two approaches in Figure 1: Holistic embedding-based and Trait embedding-based essay scoring.

In the holistic essay scoring model, the encoder takes the text of the essay as input and generates a holistic embedding, which is then used by the decoder to predict the overall score of the essay. In contrast, the core idea of trait embedding-based approaches is that each trait-specific sub-model predicts a score for its respective trait, and the average of these trait scores is computed as the holistic score, although there are variations in how the holistic score is calculated. Note that, unlike holistic embedding-based models, trait embedding-based models use trait embeddings instead of holistic embeddings.



(a) Holistic embedding-based AES



(b) Trait embeddings-based AES

Figure 1: Concepts of conventional holistic embedding and trait embeddings-based essay scoring.

From our perspective, holistic embeddings represent contextual information in an integrated form, making it difficult to capture fine-grained patterns, whereas trait embeddings encode local contextual information. In our approach, similar trait embeddings are represented as a homogeneous embedding, whereas two fundamentally different homogeneous embeddings are represented as a heterogeneous embedding. These embeddings capture global contextual information. By considering both local and global context, the proposed model understands essay texts from diverse embedding perspectives and more effectively aligns essays with their scores.

Dataset	Prompt	Dimension	Trait
ASAP ASAP++	1-2	d_1: Expression	a_1: Word choice
			a_2: Sentence fluency
			a_3: Conventions
		d_2: Organization	a_4: Organization
		d_3: Content	a_5: Content
	3-6	d_1: Expression	a_1: Narrativity
		d_2: Organization	a_2: Prompt adherence
			a_3: Language
		d_3: Content	a_4: Content
	7	d_1: Expression	a_1: Word choice and language
			a_2: Grammar and mechanics
		d_2: Organization	a_3: Organization
		d_3: Content	a_4: Content
	8	d_1: Expression	a_1: Language and style
			a_2: Grammar and mechanics
		d_2: Organization	a_3: Organization
		d_3: Content	a_4: Argument and persuasion
AIHUB	1-14	d_1: Expression	a_1: Elasticity of sentence expression
			a_2: Grammar accuracy
			a_3: Appropriateness of word use
		d_2: Organization	a_4: Appropriateness of structure within a paragraph
			a_5: Adequacy of inter-paragraph structure
			a_6: Consistency of structure
			a_7: Appropriateness of portion size
		d_3: Content	a_8: Clarity of topic
			a_9: Creativity of thought
			a_{10}: Specificity of explanation
			a_{11}: Prompt comprehension

Table 1: Essay rubric in ASAP/ASAP++ and AIHUB datasets.

In this paper, we propose a novel multi-layer encoder-based essay scoring model [1]. Given an essay as input, each trait encoder generates its trait embedding. Next, each homogeneous encoder takes similar embeddings of the traits as input and produces its homogeneous embedding. Subsequently, each heterogeneous encoder receives two homogeneous embeddings as input to generate its heterogeneous embedding. Finally, all embeddings are fed into the essay scorer to predict the holistic and trait scores.

The contributions of this paper are as follows.

- Unlike prior work on prompt-specific essay scoring, we propose a novel perspective that extends beyond holistic and trait embeddings by introducing additional type of embeddings and develop an AES model tailored to this perspective.

- The experimental results showed that the proposed model outperformed CEAES (Li and Pan, 2025), the best performing model in ASAP / ASAP++, improving QWK by 6.3%. Similarly, on AIHUB, the proposed model achieved a 15% improvement in QWK compared to multiBERT (Wang et al., 2022), the model with the highest performance.

[1]The source code for the proposed model is available at https://github.com/anonymous/anonymous.

2 Related Work

The prompt-specific holistic scoring AES has been considered as a classification (Larkey, 1998; Rudner and Liang, 2002), regression (Attali and Burstein, 2006; Phandi et al., 2015) or ranking (Chen and He, 2013) problem with manually extracted linguistic features such as style and grammar. To consider semantic meanings, latent semantic analysis (Kakkonen et al., 2005) and word embedding (Alikaniotis et al., 2016) were also applied. Cozma et al. (2018) proposed a hybrid scheme to combine string kernels with word embedding that shows better performance than the early stages of deep learning approaches (Wang et al., 2022).

Deep learning approaches, which avoid manual extraction of linguistic features, have also been used in AES. Alikaniotis et al. (2016) designed a score-specific embedding scheme (SSWE) with two-layer bidirectional LSTM. Taghipour and Ng (2016) also proposed deep learning-based AES with LSTM, RNN, and GRU. Later, a hierarchical model to utilize word and sentence level features (Dong and Zhang, 2016), attention mechanism (Dong et al., 2017), skipflow to overcome the limitations of LSTM for long sequences of sentences (Tay et al., 2017) were proposed. However, the performance of early deep learning approaches was inferior to traditional approaches with handcrafted linguistic characteristics.

Later, Mayfield and Black (2020) applied a fine-tuned pre-trained BERT language model, Yang et al. (2020) combined regression and ranking with a fine-tuned pre-trained BERT model, and Wang et al. (2022) used a hierarchical model with multiple loss functions for the token, segment and document level with pre-trained BERT model. Xie et al. (2022) used a pairwise contrastive learning to optimize regression and ranking at the same time, which shows the best performance so far. However, even with competitive performance, the aforementioned approaches using pre-trained BERT models lack explainability (Li and Ng, 2024a,b). Recently, several approaches tried to provide rationale using trait embeddings (Wang and Liu, 2025a; Chu et al., 2025; Li and Pan, 2025).

Others proposed a cross-prompt scoring scheme that separates prompts between training and testing environments to reflect realistic AES usage (Cao et al., 2020; Ridley et al., 2020, 2021; Chen and Li, 2023; Do et al., 2023; Chen and Li, 2024; Do et al., 2025), trait scoring scheme to overcome the

limitations of holistic scoring models and improve explainability (Mathias and Bhattacharyya, 2020; Ridley et al., 2021; Kumar et al., 2022a; Do et al., 2023; Chen and Li, 2023; Jiang et al., 2023; Wang and Liu, 2025b), reinforcement learning (Wang et al., 2018; Do et al., 2024b; Li and Pan, 2025) and autoregression (Do et al., 2024a,b). With the popularity of LLMs, several researchers have also explored LLMs in AES (Mizumoto and Eguchi, 2023; Han et al., 2023; Lee et al., 2024; Xiao et al., 2024; Chu et al., 2025; Song et al., 2025; Eltanbouly et al., 2025). However, the performance shows worse than that of using pre-trained BERT (Xiao et al., 2024; Mansour et al., 2024).

3 Problem Definition

Our comparison of rubrics across several essay datasets revealed that they are generally divided into three criteria: content, expression, and organization. As shown in Table 1, regardless of whether the dataset is from ASAP/ASAP++ or AIHUB, traits can be categorized into content, expression, and organization. Therefore, while the number of traits defined in each dataset may vary, the number of homogeneous and heterogeneous encoders can be aligned with the content, expression, and organization criteria.

Compared to AIHUB, ASAP/ASAP++ traits are relatively simple and lack detail. Additionally, since ASAP/ASAP++ does not provide dimensional information, we mapped similar traits to one of the three dimensions.

Essay embeddings generated by the trait encoder pre-trained on a trait are defined as **trait embeddings**. All trait embeddings within the same dimension are represented as a **homogeneous embedding**, whereas a pair of homogeneous embeddings is represented as a **heterogeneous embedding**.

4 Proposed Model

Figure 2 illustrates the proposed model, which is composed of the multi-layer encoder model and the scorer. The following sections will explain the details of each module.

4.1 Multi-layer Encoders

4.1.1 Trait Encoders

The purpose of trait encoders, $E^{(1)}_{i=\{a_i|i\in n\}}$, is to generate the i-th *trait embedding* for an essay e. Referring to the traits in Table 1, an encoder is created for each of the n traits. Such encoders are denoted by $E^{(1)}_1, E^{(1)}_2, ..., E^{(1)}_n$. Each encoder $E^{(1)}_i$ is fine-tuned using the pre-trained RoBERTa (Liu et al., 2019) as shown in Equation 1.

$$h^{(1)}_i = E^{(1)}_i(e, a_i), 1 \leq i \leq n \quad (1)$$

where $h^{(1)}_i$ is a [CLS] vector containing context information about e, focused on a trait a_i. $E^{(1)}_i$ receives e as input and predicts the essay score out of 0 to m [2]. If an error occurs between the predicted score and a_i score, it optimizes $E^{(1)}_i$ to fit a_i through error backpropagation. As a result, each trait encoder becomes specialized in predicting its associated trait score. At inference time, the fully connected layers are removed, and the fine-tuned RoBERTa then produces a trait-specific embedding for the input essay. After fine-tuning, we construct the trait encoders in Figure 2.

4.1.2 Homogeneous Encoders

The purpose of homogenenous encoder, $E^{(2)}_{i=\{d_1,d_2,d_3\}}$, is to generate the i-th *homogeneous embedding* for e. This encoder model consists of three independent encoders optimized for 'organization', 'content', and 'expression', respectively, and are indicated in order as $(E^{(2)}_1, E^{(2)}_2, E^{(2)}_3)$. Each encoder $E^{(2)}_i$ consists of a mixture of experts (MoE), consisting of 3 to 4 experts. For example, in the AIHUB dataset, the encoder $E^{(2)}_2$ specialized in 'organization' is composed of four experts $(\phi_4, \cdots, \phi_7)$ to receive [CLS] vectors $(h^{(1)}_4, \cdots, h^{(1)}_7)$ from $(E^{(1)}_4, \cdots, E^{(1)}_7)$. An expert ϕ_i consists of 5 feed-forward layers $(\phi^{(l_1)}_i, \phi^{(l_2)}_i, \cdots, \phi^{(l_5)}_i)$, and outputs $h^{(1)}_{\phi_i}$ through Equation 2.

$$h_{\phi^{(l_j)}_i} = ReLU(W_{\phi^{(l_{j-1})}_i} h_{\phi^{(l_{j-1})}_i} + b_{\phi^{(l_{j-1})}_i}) \quad (2)$$

where $h_{\phi^{(l_0)}_i} = h^{(1)}_1$ and $h^{(1)}_{\phi_i} = h_{\phi^{(l_5)}_i}$.

Suppose that $|\phi^{(l_j)}_i|$ is the number of hidden layers, 5 feed-forward layers are formed by $|\phi^{(l_3)}_i| < |\phi^{(l_2)}_i| = |\phi^{(l_4)}_i| < |\phi^{(l_1)}_i| = |\phi^{(l_5)}_i|$. As the input vector moves to the middle layer, ϕ_i removes irrelevant information from the vector and compresses it to retain only the important details. On the other hand, as the compressed vector goes from

[2] In the AIHUB dataset, all prompts share a score range of 0 to 3, while in the ASAP dataset, the range varies by prompt from 0 to 60.

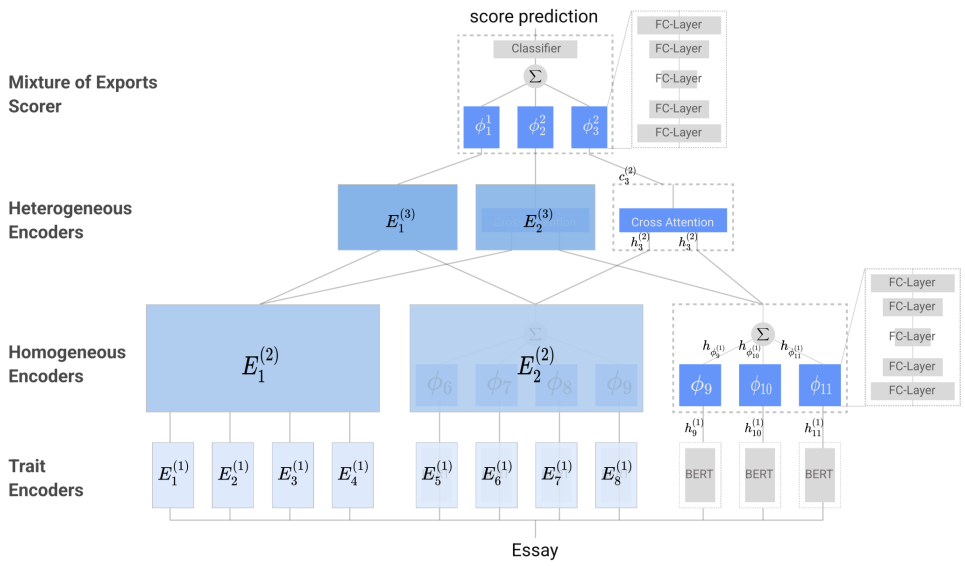


Figure 2: Network architecture of the proposed model (focusing on the AIHUB dataset for clearer explanation).

$\phi_i^{(l_3)}$ to $\phi_i^{(l_5)}$, ϕ_i learns to restore the original input vector using the compressed vector. Finally, the outputs of experts, $(h_{\phi_4}^{(1)}, \cdots, h_{\phi_7}^{(1)})$ are integrated into a single vector and their dimensions are reduced through a fully-connected layer to send to the heterogeneous encoders. Additionally, $E_i^{(2)}$ calculates the weights for the trait embeddings and the homogeneous embedding vectors.

4.1.3 Heterogeneous Encoders

The purpose of heterogeneous encoder, $E_{i=\{1,2,3\}}^{(3)}$, is to generate the i-th *heterogeneous embedding* for e. In Figure 2, $h_1^{(2)}$, $h_2^{(2)}$, and $h_3^{(2)}$ represent the homogeneous embedding vectors for 'expression', 'organization', and 'content', respectively. For each pair of homogeneous embedding vectors, Equation 3 is performed.

$$c_i^{(2)} = E_i^{(3)}(h_x^{(2)}, h_y^{(2)}). \tag{3}$$

where heterogeneous encoder $E_i^{(3)}$ (1) maps $h_x^{(2)}$ and $h_y^{(2)}$ to Query $Q = W_q h_x^{(2)}$, Key $K = W_k h_y^{(2)}$, and Value $V = W_v h_y^{(2)}$, where W_q, W_k, and W_v are learned weight matrices; (2) calculates the attention scores as $s_{xy} = \mathbf{q}_x \cdot \mathbf{k}_y$ and the attention weights as $\alpha_{xy} = \frac{exp(s_{xy})}{\Sigma_k exp(s_{xk})}$; and (3) generates the heterogeneous embedding vector $c_i^{(2)}$ between $h_x^{(2)}$ and $h_y^{(2)}$ as $c_i^{(2)} = \alpha \cdot V$.

4.2 A Mixture of Experts (MoE)-based Scorer

The purpose of scorer, S, is to predict the holistic score for e using $c_1^{(2)}$, $c_2^{(2)}$, and $c_3^{(2)}$. The scorer has the same architecture as the homogeneous encoder models, except that it operates as a single model. Specifically, it takes heterogeneous embedding vectors–$c_1^{(2)}$, $c_2^{(2)}$, and $c_3^{(2)}$–as input from heterogeneous encoders and predicts the essay score on a scale from 0 to m. Additionally, it calculates the weights for the heterogeneous embedding vectors.

4.3 Loss Function

In the i-th trait encoder, the cross-entropy error is computed by $L_{CE} = -\sum_{i=0}^{n} s_{a_i} \log \hat{s}_{a_i}$, where s_{a_i} and $\hat{s}_{a_i}$ represent the ground truth trait score and the encoder's predicted trait score for the i-th essay, e_i, in the training set. Similarly, in the entire model, the cross-entropy error is computed by $L_{CE} = -\sum_{i=0}^{n} s_i \log \hat{s}_i$, where s_i and $\hat{s}_i$ represent the ground truth holistic score and the model's predicted score for the i-th essay, e_i, in the training set. We also utilize contrastive learning optimized for the automated essay scoring task; see Appendix 4.4 for details.

4.4 Contrastive Learning for Automated Essay Scoring

In (Yu et al., 2023), we proposed a novel contrastive learning method, which differs from conventional contrastive learning that treats two texts as a positive pair if their content is identical, and as a negative pair if it differs. This encourages embeddings of similar texts to be close in the representation space, while dissimilar ones are pushed apart. In contrast, essay scoring requires a different notion

of similarity. For instance, an essay scored 3 is not similar to another essay with identical content, but rather to the shared characteristics – such as grammar, word choice, and organization –typically observed in essays with the same score. Accordingly, the proposed method defines the positive sample as the mean embedding of essays with a score of 3, and the negative sample as the mean embedding of essays with a randomly chosen score other than 3.

In Equation 4, e_i represents the embedding of the input data, while e_p and e_n refer to the positive and negative samples, respectively. $dist$ denotes the Euclidean distance, and sim represents the cosine similarity. λ_1 and λ_2 are hyperparameters, set to 5 and 10 in all experiments.

$$\begin{aligned} L_{CL} &= \lambda_1 \cdot p + \lambda_2 \cdot q \\ p &= \frac{dist(e_i, e_p)}{\{dist(e_i, e_p) + dist(e_i, e_n)\}} \\ q &= |sim(e_i, e_n)| \end{aligned} \tag{4}$$

The contrastive loss function includes two terms, p and q. The term p encourages the embedding of the input essay to be close to that of the positive sample (i.e., minimizing the distance), while q encourages it to be distant from the negative sample (i.e., maximizing the distance up to a value of 1).

The total loss is defined as follows:

$$L_{total} = \gamma \cdot L_{CE} + (1 - \gamma) \cdot L_{CL} \tag{5}$$

, where γ is a hyperparameter set to 0.5.

4.5 Inference Step

In this step, a new essay is given as input, the proposed model predicts the essay score from 0 to m and calculates the weights for n traits, 3 homogeneous embeddings, and 3 heterogeneous embeddings through a softmax layer. For example, as shown in Figure 2, the heterogeneous encoder $E_1^{(2)}$ calculates $(w_{\phi_1}^{(1)}, \cdots, w_{\phi_4}^{(1)})$, the weight values corresponding to $(h_{\phi_1}^{(1)}, \cdots, h_{\phi_4}^{(1)})$. The scorer calculates $(w_{\phi_1}^{(2)}, w_{\phi_2}^{(2)}, w_{\phi_3}^{(2)})$, which are the weight values corresponding to $(c_1^{(2)}, c_2^{(2)}, c_3^{(2)})$.

4.6 Complexity Analysis of Proposed Model

To analyze the time complexity of the proposed model, we divide it into four main stages and calculate the total number of parameters required for training in each stage. Based on our detailed analysis, the number of parameters in the trait encoders, the homogeneous encoders, the heterogeneous encoders, and the scorer are $1,639M$, $18M$, $20M$, and $6M$, respectively. Thus, the total number of parameters in the proposed model is approximately $1,683M$. For further details, refer to Appendix A.

5 Experimental Setup

To evaluate the performance of the proposed method, we used two datasets. The first is well known ASAP/ASAP++ (Alves et al., 2015; Mathias and Bhattacharyya, 2018) and the second is AIHUB [3] dataset. Tables 8 and 9 in Appendix B show the characteristics of ASAP/ASAP++ and AIHUB essay data, respectively. The ASAP/ASAP++ dataset consists of approximately 13,000 essays in 8 prompts. Each essay was evaluated by two evaluators and the overall scores are available from prompt 1 to prompt 8. The AIHUB essay data have approximately 50,413 essays with various rubrics as shown in Table 1. The AIHUB essay data cover 5 different essay styles, 14 topics from 5th to 12th graders. Each essay was evaluated by three experts with evaluation rubrics with 11 criteria. The latter is known to be more difficult to predict the score, since it includes diverse styles of different prompt essays written by varying ages of students. In addition, the average length of the essay is 141 words, making it more difficult to predict the score. For details, see Appendix B.

The experiment was executed on a single machine equipped with the NVIDIA GPU (GeForce A6000) with 500GB SSD and 128G main memory. We used Python (v3.8) and PyTorch(v1.11) with CUDA (v11.3). Each experiment was conducted for 200 epochs with a dropout rate of 0.2, using the Adam optimizer with a pre-trained RoBERTa, a learning rate of 10^{-5}, and a batch size of 128.

6 Experimental Results

6.1 Evaluation of Prompt-specific Holistic Scoring

Table 2 shows the experimental results for the ASAP/ASAP++ dataset. The hierarchical approaches using BERT from Cao et al. (2020), R2BERT (Yang et al., 2020), Wang et al. (2022),

[3] Datasets from 'The Open AI Dataset Project (AIHUB, S. Korea)'. Data information can be accessed through 'AIHUB (www.aihub.or.kr)'.

Approach	Model	P1	P2	P3	P4	P5	P6	P7	P8	Avg.
Holistic embedding	EASE(SVR) (Phandi et al., 2015)	0.781	0.621	0.630	0.749	0.782	0.771	0.727	0.534	0.699
	EASE(BLRR) (Phandi et al., 2015)	0.761	0.606	0.621	0.742	0.784	0.775	0.730	0.617	0.705
	CNN+LSTM (Taghipour and Ng, 2016)	0.821	0.688	0.694	0.805	0.807	0.819	0.808	0.644	0.761
	LSTM-CNN-Attn (Dong et al., 2017)	0.822	0.682	0.674	0.814	0.803	0.811	0.801	0.705	0.764
	Skipflow(Bilinear) (Tay et al., 2017)	0.830	0.678	0.677	0.778	0.795	0.807	0.790	0.670	0.753
	Skipflow(Tensor) (Tay et al., 2017)	0.832	0.684	0.695	0.788	0.815	0.810	0.800	0.697	0.764
	Cross(HA-LSTM+SST+DAT) (Cao et al., 2020)	0.836	0.730	0.732	0.822	0.835	0.832	0.821	0.718	0.790
	Cross(BERT+SST+DAT) (Cao et al., 2020)	0.824	0.699	0.726	0.859	0.822	0.828	0.840	0.726	0.791
	R2BERT (Yang et al., 2020)	0.817	0.719	0.698	0.845	0.841	0.847	0.839	0.744	0.794
	multiBERT(BERT-DOC-TOK-SEG) (Wang et al., 2022)	0.836	0.695	0.700	0.815	0.812	0.816	0.838	0.744	0.782
	multiBERT(Tran-BERT-MS-ML-R) (Wang et al., 2022)	0.834	0.716	0.714	0.812	0.813	0.836	0.839	0.766	0.791
	NPCR (Xie et al., 2022)	0.856	0.750	0.756	0.851	0.847	0.858	0.838	0.779	0.817
	Fine-tuned GPT-3.5 (Xiao et al., 2024)	0.741	0.618	0.704	0.859	0.796	0.848	0.727	0.613	0.738
Trait embeddings	MTL-BiLSTM (Kumar et al., 2022b)	0.831	0.689	0.687	0.798	0.800	0.813	0.795	0.699	0.764
	STL-LSTM (Kumar et al., 2022b)	0.818	0.658	0.653	0.780	0.789	0.803	0.786	0.697	0.748
	BERT-STL (Kumar et al., 2022b)	0.800	0.679	0.679	0.822	0.803	0.797	0.827	0.725	0.767
	ArTS-Large (Do et al., 2024a)	0.700	0.699	0.704	0.766	0.725	0.770	0.739	0.644	0.718
	SaMRL-Large (Do et al., 2024b)	0.702	0.711	0.708	0.766	0.722	0.773	0.743	0.649	0.722
	RMTS-T5 (Chu et al., 2025)	0.716	0.704	0.723	0.772	0.737	0.769	0.736	0.651	0.726
	T-MES (Wang and Liu, 2025b)	0.728	0.684	0.702	0.771	0.726	0.754	0.755	0.629	0.719
	CEAES (Li and Pan, 2025)	0.863	0.773	0.759	0.865	0.842	0.856	0.843	0.765	0.821
Multi-layer embeddings	**Proposed model**	**0.886**	**0.856**	**0.827**	**0.903**	**0.882**	**0.859**	**0.899**	**0.870**	**0.873**

Table 2: Experimental results of prompt-specific holistic scoring models using the ASAP/ASAP++ dataset in terms of QWK.

Approach	Model	C1	C2	C3	C4	C5	Avg.
Holistic embedding	EASE(SVR) (Phandi et al., 2015)	0.601	0.311	0.328	0.319	0.109	0.386
	EASE(BLRR) (Phandi et al., 2015)	0.612	0.339	0.328	0.340	0.142	0.404
	CNN+LSTM (Taghipour and Ng, 2016)	0.642	0.438	0.349	0.373	0.162	0.439
	LSTM-CNN-Attn (Dong et al., 2017)	0.646	0.454	0.357	0.411	0.189	0.456
	Skipflow(Bilinear) (Tay et al., 2017)	0.630	0.451	0.361	0.473	0.192	0.465
	Skipflow(Tensor) (Tay et al., 2017)	0.634	0.455	0.413	0.471	0.192	0.474
	R2BERT (Yang et al., 2020)	0.669	0.452	0.441	0.496	0.402	0.531
	multiBERT(BERT-DOC-TOK-SEG) (Wang et al., 2022)	0.725	0.435	0.452	0.751	0.725	0.663
	multiBERT(Tran-BERT-MS-ML-R) (Wang et al., 2022)	0.737	0.461	0.504	0.773	0.741	0.683
	NPCR (Xie et al., 2022)	0.729	0.592	0.584	0.501	0.477	0.601
	Fine-tuned GPT-3.5 (Xiao et al., 2024)	0.731	0.302	0.328	0.406	0.343	0.493
Trait embeddings	MTL-BiLSTM (Kumar et al., 2022b)	0.535	0.454	0.448	0.518	0.465	0.477
	STL-LSTM (Kumar et al., 2022b)	0.557	0.499	0.462	0.549	0.471	0.498
	BERT-STL (Kumar et al., 2022b)	0.590	0.370	0.530	0.520	0.450	0.499
	ArTS-Large (Do et al., 2024a)	0.573	0.368	0.291	0.678	0.694	0.521
	RMTS-T5 (Chu et al., 2025)	0.580	0.373	0.294	0.687	0.702	0.527
	T-MES (Wang and Liu, 2025b)	0.468	0.203	0.258	0.522	0.517	0.376
Multi-layer embeddings	**Proposed model**	**0.829**	**0.667**	**0.656**	**0.823**	**0.832**	**0.787**

Table 3: Experimental results of prompt-specific holistic scoring models using the AIHUB dataset in terms of QWK. Categories C1–C5 represent narrative essays, writing tasks, alternative proposals, argumentative essays, and agree/disagree tasks, respectively. AES models whose source code was not publicly available were excluded from the experiment.

and NCPR (Xie et al., 2022) show the best performance. Despite the growing popularity of LLMs, fine-tuned models like GPT-3.5 still perform worse than BERT-based models. The performance of recently proposed trait-based models, such as MTL-BiLSTM (Kumar et al., 2022b) and BERT-STL (Kumar et al., 2022b), is comparable to that of BERT-based holistic scoring models like R2BERT (Yang et al., 2020) and NCPR (Xie et al., 2022). The BERT-STL model takes an essay as input and uses BERT to generate trait-specific embeddings, which are then used to predict the score for a single target trait. The MTL-BiLSTM model uses a multi-task learning framework where each trait is predicted by a dedicated BiLSTM. Trait embeddings are generated and used to compute individual scores that are then linearly combined into a holistic score. The BiLSTM captures contextual patterns using word and sentence embeddings, along with word- and sentence-level attention. T-MES (Wang and Liu, 2025b) uses a Transformer encoder to embed essay words and utilizes the weights of a Mixture-of-Experts (MoE) model to aggregate these word embeddings into trait embeddings. It further applies contrastive learning to bring similar trait embeddings closer while pushing dissimilar ones apart. However, since this model generates trait embeddings solely from word embeddings, its overall performance remains limited. Our proposed model shows the best performance as in **0.873** QWK, indicating an improvement of approximately 7% in performance, compared to NPCR on the ASAP/ASAP++ dataset. Recent sequence based models (ArTS (Do et al., 2024a) and RMTS (Chu et al., 2025)) show competitive performance with pre-trained T5. So far, CEAES (Li and Pan, 2025), optimizing the score and feedback using reinforcement learning, shows the best performance with 0.821 in QWK which is still inferior to the proposed model.

The performance of the AIHUB dataset is shown in Table 3. The proposed model demonstrates the best performance with a QWK of **0.787**, indicating an improvement of approximately 15% compared to multiBERT(Tran-BERT-MS-ML-R), which has shown the best performance on the AIHUB dataset so far. Compared to the ASAP/ASAP++ dataset, the overall QWK on the AIHUB dataset is lower due to a significant imbalance in the number of essays across score ranges. Apart from the proposed model, NPCR [4] outperforms on the ASAP/ASAP++ dataset, while multiBERT [5] performs better on the AIHUB dataset. Depending on the characteristics of the data, contrastive learning or multi-level essay representation learning appears to have a slight advantage in performance.

In summary, the experimental results in Tables 2 and 3 clearly show that the significant performance improvement lies in the ability of the proposed model to effectively leverage various types of embeddings, i.e., trait, homogeneous, and heterogeneous embeddings. By generating trait embeddings that capture the contextual information of the entire text through rubrics in Table 1 and further exploiting homogeneous and heterogeneous embeddings, the proposed model can hierarchically capture the context of an essay, leading to more accurate predictions of its holistic score.

6.2 Evaluation of Prompt-specific Trait Scoring

Table 4 summarizes the QWK scores of the prompt-specific trait scoring models. In ASAP/ASAP++, there are Cont, Org, WC, SF, Conv, PA, Lang, and Nar, which represent content, organization, word count, style and fluency, convention, prompt adherence, language, and narration, respectively. In AIHUB, 11 traits (a_1-a_{11}) are grouped into three categories such as Expression, Organization, and Contents. See Table 1 for details of traits.

In ASAP/ASAP++ dataset, the proposed model achieves the best performance, followed by RMTS among the trait-based models. However, in AIHUB dataset, the proposed model shows slightly better performance than others. Surprisingly, ArTS shows the better performance than T-MES and RMTS. Considering that RMTS extract trait specific rationales and then applied the rationales in the essay scoring system, we conjecture that the hierarchical rubrics and korean language characteristics prohibits RMTS to properly extract the prompt-specific rationales at the beginning and drags the performance of RMTS on AIHUB dataset. On the other hands, the sequence based ArTS are not influenced by korean language characteristics.

In T-MES, trait embeddings are generated solely from word embeddings, which results in strong performance primarily on traits that capture broad or overall narrative aspects, such as grammar errors or narration. In contrast, the proposed model demonstrates higher performance with traits related to more specific content, including appropriate word usage, organization, creativity, and the specificity of explanations. This difference arises because the proposed model generates trait embeddings, using trait encoders $E^{(1)}$, which incorporate the full contextual information of the essay text. Overall, both STL-LSTM and BERT-STL models exhibit low performance across most traits. STL-LSTM generates trait embeddings using word and sentence embeddings, while BERT-STL utilizes word, sentence, and text embeddings. In the ASAP/ASAP++ dataset, there appears to be a stronger correlation between vocabulary and prompts compared to the AIHUB dataset that likely explains why the PA trait shows slightly better performance for these models compared to T-MES and the proposed model.

6.3 Ablation Study

Table 5 shows the QWK values of the ablated models: trait embedding-based scoring (M1), trait+homogeneous embedding-based scoring (M2), trait+homogeneous+heterogeneous embedding-based scoring (M3), and trait+homogeneous+heterogeneous embedding-based scoring with contrastive learning. The experimental results show that the performance follows the order of M4, M3, M2, and M1, indicating that scoring with diverse embeddings is a key factor in performance improvement. Interestingly, incorporating heterogeneous embeddings (i.e. from M2 to M3) led to a dramatic increase in performance. The reason can be discussed in detail in Section 6.4.

[4] NPCR performs contrastive learning to group similar essays closely together while placing dissimilar essays farther apart, coupled with a ranking technique to predict relative scores between essays and reference essays.

[5] multiBERT utilizes multi-level essay representation such as spanning words, segments, and documents with a multi-task loss function considering both regression and ranking.

Dataset	Trait	MTL-BiLSTM	STL-LSTM	BERT-STL	T-MES	ArTS	RMTS	Proposed model
ASAP	Cont	0.685	0.707	0.719	0.712	0.754	0.737	**0.762**
ASAP++	Org	0.615	0.649	0.669	0.667	0.672	0.682	**0.735**
	WC	0.615	0.621	0.676	0.693	0.679	0.705	**0.734**
	SF	0.598	0.612	0.625	0.669	0.678	0.694	**0.732**
	Conv	0.560	0.605	0.656	0.673	0.668	0.690	**0.694**
	PA	0.701	0.731	0.731	0.719	0.751	0.752	**0.764**
	Lang	0.604	0.640	0.659	0.662	0.698	**0.713**	0.682
	Nar	0.668	0.699	0.703	0.716	0.725	0.744	**0.755**
AIHUB	a_1	0.132	0.171	0.183	0.207	0.180	0.193	**0.248**
	a_2	0.241	0.224	0.235	**0.807**	0.250	0.261	0.338
	a_3	0.718	0.738	0.771	0.603	0.787	**0.800**	0.758
	a_4	0.122	0.110	0.213	**0.418**	0.235	0.247	0.242
	a_5	0.412	0.405	0.468	0.205	0.813	**0.825**	0.746
	a_6	0.571	0.611	0.636	0.113	0.708	**0.722**	0.703
	a_7	0.186	0.244	0.276	0.356	0.472	0.469	**0.491**
	a_8	0.114	0.018	0.130	0.281	0.235	0.247	**0.310**
	a_9	0.518	0.489	0.505	0.162	0.139	0.152	**0.541**
	a_{10}	0.593	0.637	0.682	0.224	0.820	**0.833**	0.775
	a_{11}	0.332	0.295	0.318	**0.950**	0.294	0.304	0.358

Table 4: Experimental results of prompt-specific trait scoring models in terms of QWK.

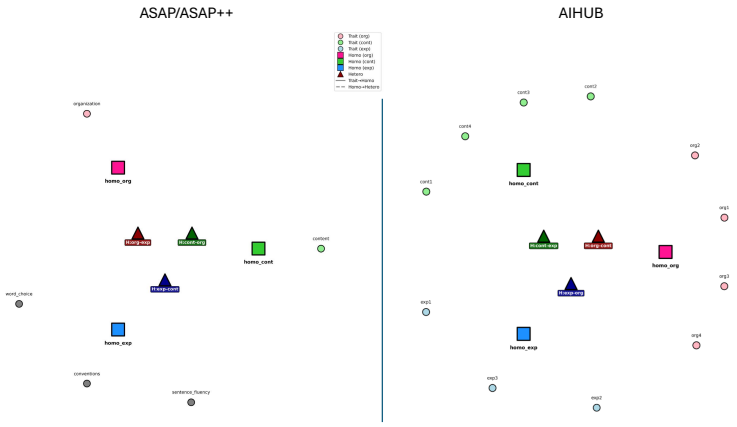


Figure 3: Visualization results of trait, homogeneous, and heterogeneous embeddings.

Dataset	Prompt	M1	M2	M3	M4
ASAP	P1	0.602	0.646	0.810	**0.886**
ASAP++	P2	0.585	0.601	0.792	**0.856**
	P3	0.759	0.775	0.806	**0.827**
	P4	0.753	0.787	0.882	**0.903**
	P5	0.720	0.744	0.866	**0.882**
	P6	0.745	0.769	0.839	**0.859**
	P7	0.694	0.735	0.869	**0.899**
	P8	0.732	0.758	0.841	**0.870**
	Avg.	0.699	0.727	0.838	**0.873**
Dataset	**Category**	**M1**	**M2**	**M3**	**M4**
AIHUB	C1	0.766	0.782	0.811	**0.829**
	C2	0.591	0.622	0.651	**0.667**
	C3	0.571	0.584	0.629	**0.656**
	C4	0.616	0.638	0.818	**0.823**
	C5	0.641	0.669	0.824	**0.832**
	Avg.	0.666	0.686	0.772	**0.787**

Table 5: Ablation study of the proposed model in terms of QWK.

6.4 Visualization-driven Insights into Experimental Results

Figure 3 presents the visualization results of different types of embedding vectors of a sample essay on radial hierarchical embeddings. Left shows a sample from ASAP/ASAP++ dataset, and the right shows a sample from AIHUB dataset. The traits are grouped around the homogeneous embeddings (homo_cont, homo_org, homo_expr), and the homogeneous embeddings are grouped around the heterogeneous embeddings (H:cont-exp, H:org-cont, H:exp-org).

6.5 Weights of Multi-layer Embeddings

Table 6 shows the results of the proposed model for the weights of the trait, homogeneous, and heterogeneous embeddings of an example essay (i.e. Essay 19 from Appendix C). Essay 19 corresponds to the C5 category in the AIHUB dataset and is about the author's opinion on whether to support or oppose the internet real-name system.

The most important traits are a_2, a_3, a_6, and

Embedding	$\hat{s_{a_i}}$	s_{a_i}	Weight w
a_1	2	2	0.0818
a_2	3	3	**0.1003**
a_3	2	3	**0.1920**
a_4	3	2	0.0610
a_5	3	3	0.0716
a_6	2	3	**0.1137**
a_7	2	2	**0.1115**
a_8	3	3	0.0428
a_9	2	2	0.0995
a_{10}	2	3	0.0493
a_{11}	3	3	0.0764
d_1			**0.3741**
d_2			**0.3578**
d_3			0.2680
d_1-d_2			**0.9933**
d_1-d_3			0.0059
d_2-d_3			0.0009
$\hat{s_i}$	2	s_i	3

Table 6: Weights of Essay $i = 19$. $\hat{s}_{a_i}$, s_{a_i}, $\hat{s}_i$, and s_i represent the predicted score for trait a_i, the true score for a_i, the predicted holistic score for essay e_i, and the true holistic score for e_i, respectively.

a_7, which correspond to 'd_1: expression' and 'd_2: organization', and have the greatest impact on predicting the holistic score of Essay 19. Similarly, the most important homogeneous embeddings are d_1 and d_2, and the most important heterogeneous embedding is d_1-d_2. This suggests that the weights of the most important traits are correlated with both homogeneous and heterogeneous embeddings. Interestingly, the traits related to 'd_3: content' in this essay have low weights and exert little influence on predicting the holistic score. Referring to Table 12, the feedback from the expert who evaluated this essay suggests that the expert also pointed out shortcomings in the traits related to 'd_3: content.' Therefore, it can be inferred that the expert's feedback is correlated with the weights in the table. The weight values for traits, homogeneous modalities, and heterogeneous modalities make it to be helpful to understand the underlying scoring rationale of a target essay.

7 Conclusion

To enhance holistic scoring models, we propose a multi-layer encoder approach that produces various embeddings with corresponding weights. Experiments show that the proposed model improves overall QWK by up to 7% on ASAP/ASAP++ and 15% on AIHub essay scoring datasets. Furthermore, the experimental results demonstrate that our approach helps to understand the reasons behind the holistic score, and can be used to generate feedback.

Limitations

Three main limitations of the proposed approach are inherited from the dataset and evaluation metrics. The ASAP dataset is well known and widely used in AES community. However, the dataset size is way too small and includes inherited bias from human raters. The AIHUB has similar drawbacks from size and inherited bias. The second drawback is the evaluation metric. The performance of a model was measured by QWK which is the most popular in academic community. However, QWK does not separately measure the relevance to prompt and quality of the essay. In this paper, we are trying to overcome the latter by using multiple traits but used QWK as a performance indicator. The third limitation is the model complexity and memory usage. Since the proposed model utilized multiple encoders with hierarchy, it requires more memory usage (Details are analyzed in the Appendix). Considering the number of parameters of recent LLMs, it requires much less resources. In the future, we are planning to devise a scheme to reduce the number of parameters with similar performance.

Ethics Statement

The proposed method does not consider any ethical aspects of the student essay dataset. However, the algorithm can be effected by the training data without any notice. By collecting more dataset from diverse ethic groups, we can avoid potential risks in ethics.

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A Complexity Analysis of Proposed Model

To analyze the time complexity of the proposed model, we divide the model into four main stages and calculate the total parameters required for training at each stage. Table 7 shows the notation terms for discussing the complexity of the proposed model.

A.1 Trait Encoder

There are a total of 11 encoders based on the AIHUB dataset. Each encoder is a RoBERTa [6], which consists of three components: the embedding layer, the transformer block, and the output layer.

- The token embedding $P_{te} = V \cdot H = 50,265 \cdot 768 \approx 39M$ and the positional embedding $P_{pe} = S \cdot H = 4,096 \cdot 768 \approx 3M$. The total number of parameters is $42M$.

- A transformer block consists of a self-attention, a feed-forward network, and layer normalization. The number of weights for Query, Key, Value, and Output is $4 \cdot (H \cdot H) = 4 \cdot (768 \cdot 768) \approx 2M$. Since the feed-forward network is composed of two layers, the number of parameters is $H \cdot F + F \cdot H = 2 \cdot (768, 3,072) \approx 5M$. The number of parameters in LayerNorm is $2 \cdot H = 2 \cdot 768 \approx 2M$. As a result, the total number of parameters per transformer block is approximately $7M$. Since there are 12 transformer blocks, the total number of parameters is $L \cdot 7M = 12 \cdot 7M \approx 85M$.

- The number of parameters in the output layer is $V \cdot H = 50,265 \cdot 768 = 39M$.

In conclusion, the number of parameters in a single encoder is approximately $149M$. Finally, since there are 11 encoders, the total number of parameters is $11 \cdot 149M \approx 1,639M$.

A.2 Homogeneous Encoder

As shown in Figure 2, there are three encoders representing 'organization', 'expression', and 'content'. The first encoder consists of four experts and

[6] The input token length for BERT is limited to a maximum of 512 tokens. In the ASAP and AIHUB datasets, the length of many essays exceeds this limit. For reference, essays with fewer than 512 tokens represent 36%, while essays with 512 tokens or more represent 64%. The maximum token count for an essay is 3,071. For this reason, RoBERTa is used in this study.

Model	Term	Description	Value
Trait Encoder	H	Hidden size	768
	L	Number of layers	12
	A	Number of attention heads	12
	F	Feed-forward size	3,072
	V	Vocabulary size	50,265
	S	Maximum input sequence length	4,096
Homogeneous Encoder, Scorer	H	Hidden size	768
	L	Number of feed-forward layers	5
	F	Number of hidden nodes	$F_{l_1} = F_{l_5} = 768$ $F_{l_2} = F_{l_4} = \frac{H}{2}$ $F_{l_3} = \frac{H}{4}$
	$N_{experts}$	Number of experts per decoder	4
	$D_{decoders}$	Number of decoders	3
	d_{input}	input vector dimension	768
Heterogeneous Encoder	H	Hidden size	768
	W_Q	Query weight matrices	$H \cdot H$
	W_K	Key weight matrices	$H \cdot H$
	W_v	Value weight matrices	$H \cdot H$
	b_Q	Query bias	H
	b_K	Key bias	H
	b_V	Value bias	H

Table 7: Notation terms.

Prompt	Genre	#Essays	Length	Range
1	Arg.	1,783	350	2-12
2	Arg.	1,800	350	1-6
3	Res.	1,726	150	0-3
4	Res.	1,772	150	0-3
5	Res.	1,805	150	0-4
6	Res.	1,800	150	0-4
7	Res.	1,569	250	0-30
8	Res.	723	650	0-60

Table 8: ASAP dataset. Notation. Arg.: Argumentative, Res.:Response, Length: Average Length.

a fully connected layer, the second encoder also comprises four experts and a fully connected layer, and the third decoder includes three experts and a fully connected layer.

Each expert consists of five feed-forward layers. The number of parameters for each layer is calculated as follows:

- $P_{l_1} = H \cdot F_{l_1} + F_{l_1} = H \cdot 768 + 768 \approx 0.6M$.
- $P_{l_2} = F_{l_1} \cdot F_{l_2} + F_{l_2} = 768 \cdot \frac{H}{2} + \frac{H}{2} \approx 0.3M$.
- $P_{l_3} = F_{l_2} \cdot F_{l_3} + F_{l_3} = \frac{H}{2} \cdot \frac{H}{4} + \frac{H}{4} \approx 0.07M$.
- $P_{l_4} = F_{l_3} \cdot F_{l_2} + F_{l_2} = \frac{H}{4} \cdot \frac{H}{2} + \frac{H}{2} \approx 0.07M$.
- $P_{l_5} = F_{l_2} \cdot H + H = \frac{H}{2} \cdot H + H \approx 0.3M$.

Therefore, the total number of parameters from the first to the fifth layer is $\Sigma_{i=1}^{5} P_{l_i} \approx 1.3M$.

The number of parameters in the fully-connected layer, which integrates the outputs of the experts (4H) into a single vector (H), is calculated as $P_{fc} = 4H \cdot H + H \approx 2M$.

The total number of parameters for the two decoders, each consisting of 4 experts, is $2 \cdot 4 \cdot 1.3M + 2M = 12M$, while the total number of parameters for the single encoder with 3 experts is $1 \cdot 3 \cdot 1.3M + 2M = 6M$. In conclusion, the total number of all parameters in three decoders is $18M$.

A.3 Heterogeneous Encoder

Each cross-encoder consists of a cross-attention layer and a feed-forward network layer.

Referring to Table 7, the total number of parameters in the cross-attention layer is $P_{\text{cross-attention}} = P_{W_Q} + P_{W_K} + P_{W_V} + P_{b_Q} + P_{b_K} + P_{b_V} = 3H^2 + 3H = 1.8M$.

The feed-forward network layer consists of two layers. The total parameters in the first layer are $P_{FFN_1} = H \cdot 4H + 4H = 768 \cdot 3,072 + 3,072 = 2.4M$, and in the second layer, $P_{FFN_2} = 4H \cdot H + H = 3,072 \cdot 768 + 768 = 2.4M$. Thus, the total number of parameters in the feed-forward network layer is $P_{FFN} = P_{FFN_1} + P_{FFN_2} = 2.4M + 2.4M = 4.8M$.

The number of parameters in a single cross-encoder is $P_{\text{cross-encoder}} = P_{\text{cross-attention}} + P_{FFN} = 1.8M + 4.8M = 6.6M$. Since there are three cross-encoders, the total number of parameters is $3 \cdot 6.6M = 19.8M$.

A.4 Scorer

The scorer is a single decoder with the same architecture as the homogeneous encoder. Specifically,

School	Grade	# of essays	Average # of word tokens
Elementary	4th	5,827	270
	5th	6,761	303
	6th	6,663	384
Middle	1st	6,955	352
	2nd	6,649	436
	3rd	6,189	523
High	1st	4,812	578
	2nd	4,800	637
	3rd	1,757	608
Total		50,413	455

Table 9: Statistics of the AIHUB dataset.

Category	# of prompts	# of essays	Score range
Writing tasks	3	5,506	0-3
Alternative proposals	3	7,005	0-3
Narrative essays	3	18,275	0-3
Argumentative essays	3	11,172	0-3
Agree/Disagree tasks	2	8,455	0-3
Total	14	50,413	0-3

Table 10: Five categories in the AIHUB dataset.

the encoder comprises three experts and a fully connected layer. According to Subsection A.2, the total number of parameters is calculated as $1 \cdot 3 \cdot 1.3M + 2M = 6M$.

B Essay Corpora

B.1 ASAP

Table 8 shows the characteristics of the ASAP and AIHUB essay data, respectively. The ASAP dataset consists of around 13,000 essays in 8 prompts. Each essay was evaluated by two evaluators, and overall scores are available from prompt 1 to prompt 8.

B.2 AIHUB

To make artificial intelligence technology easily accessible to everyone, the National Information Society Agency (NIA), a public institution in Korea, collected a benchmark dataset for Automated Essay Scoring (AES) and released it to the public in 2022. In this paper, we refer to such a dataset as the AIHUB dataset, which includes essays written in Korean by students from 4th to 6th grade in elementary school, 1st to 3rd grade in middle and high school, whose native language is Korean.

The total number of essays is 50,413. Table 9 summarizes basic statistics about the AIHUB dataset. Unlike the ASAP dataset, each essay in the AIHUB dataset is categorized into one of five categories: "writing," "alternative proposal," "narrative essay," "argumentative essay," and "agree/disagree." Table 10 shows the number of prompts, the total number of essays, and the range of essay scores in each category. Table 11 shows the list of essay prompts in the AIHUB dataset.

Each essay was evaluated by three experts with evaluation rubrics with 11 criteria. The latter is known as more difficult to predict a score since it includes diverse styles of different prompt essays written by varying ages of students. In addition, the average length of essay is 141 words, which make it more difficult to predict the score.

C An Essay Example

As an example of essays in AIHUB, Table 12 shows an essay that contains the essay text, the feedback from a human expert, and the feedback from GPT-4. In addition, it shows the weight values of the trait embeddings, homogeneous embeddings, and heterogeneous embeddings that are calculated by the proposed model.

Category	No	Prompt
Writing tasks	1	My biography
	2	An experience of losing something precious
	3	My thoughts on the universe
Alternative proposals	1	Solutions to gender conflicts
	2	Strategies for improving prejudice against people with disabilities
	3	My thoughts on the issue of constructing hate facilities
Narrative essays	1	My effort for my career path
	2	Movie and book review
	3	An example of when I didn't respect others
Argumentative essays	1	My thoughts on racial discrimination
	2	My thoughts on intellectual property rights
	3	My opinion on evaluation
Agree/Disagree tasks	1	My thoughts on singlehood
	2	My thoughts on anonymity

Table 11: Essay prompts corresponding per category in the AIHUB dataset.

Key	Value
ID	19
category	C5
text	The restrictive Internet real-name system infringes on freedom, as anonymity plays a crucial role in democracy. [...] Anonymous expression facilitates public opinion formation and information sharing, making the protection of free speech essential. Unlike a full real-name system, a limited identification system requires users to verify their identity with service providers while keeping their information hidden from the public. [...] However, this system still restricts free expression, functions as pre-screening, and raises privacy concerns under Article 17 of the Constitution. Additionally, storing personal data with institutions vulnerable to breaches increases the risk of identity theft. Rather than enforcing a real-name system, fostering responsibility among users is key. [...] Excessive abuse, personal attacks, and defamation can already be investigated using stored communication records under existing laws. While tracking illegal posts deters harmful content, it may also suppress lawful speech, limiting political, social, and corporate criticism. [...] A real-name system could ultimately silence free expression rather than promote accountability.

Human Feedback

Trait	Text	Score
a_1	Grammatical errors are sometimes seen	2
a_2	Use appropriate words for situations	3
a_3	Most of the sentence expressions are accurate, but sometimes there are ambiguous sentences	2
a_4	Topic and supportive sentences are closely linked	3
a_5	in the form of an introduction, body, and conclusion	3
a_6	The flow of text is generally seamlessly connected	2
a_7	Write more than ±10% different from the suggested amount	2
a_8	Subject clearly revealed	3
a_9	Most explanations are specific	2
a_{10}	There are signs of thinking about the problem, but no in-depth insights	2
a_{11}	Complete exactly what was presented at the prompt	3
Overall	Sometimes grammatical errors are seen.Properly use words for the situation.Most of the sentences are correct, but sometimes there are ambiguous sentences.The topic and backing sentences are closely connected.It has a form of introduction, main and conclusion.The flow of the article is generally connected smoothly.Writing more than ± 10% of the quantity presented.The topic is clearly revealed.Most explanations are specific.There is a trace of thinking about the problem, but there is no in -depth insight.Write exactly what we presented in the prompt.It is a good idea to consistently insist on your own argument that opposes the limited Internet real name system.However, it is a pity that the reason is somewhat common, and if you think about the argument of the opposite position, and add the objection to it, it will be a much deeper article.	2

LLM Feedback

Text	Score
Grammar & Word Usage: There's no big grammatical error, but I see a lot of word choices. Utilizing more diverse vocabulary can enrich your thesis. Sentence Clarity & Expression: The clarity and expressiveness of the sentence need to be improved a little bit. Efforts are needed to diversify the sentence structure and make the flow of writing smoother. Organization & Structure: The structure within paragraphs is consistent, but the structure between paragraphs appears to be inconsistent. It is important to have a clearer introduction, body, and conclusion structure and to strengthen the connection between paragraphs. Content & Argumentation: The clarity and concreteness of the subject need to be improved. Efforts should be made to develop arguments in more detail and detail, and to logically develop creative thinking. Overall Impression: The structure and theme of the text needs some improvement, but the perspective on freedom of expression and the real-name system is clearly communicated. If you strengthen your thesis and improve your structure through efforts, you will be highly evaluated. Keep up the good work!	1

Model Output

Key	Weight	Actual	Predicted
holistic	–	2	1
a_1, a_2, a_3	0.3741 (0.0818, 0.1003, 0.192)	(2, 3, 2)	(3, 3, 3)
a_4, a_5, a_6, a_7	0.3578 (0.061, 0.0716, 0.1137, 0.1115)	(3, 3, 2, 2)	(3, 3, 3, 3)
a_8, a_9, a_{10}, a_{11}	0.2681 (0.0428, 0.0995, 0.0493, 0.0764)	(3, 2, 2, 3)	(3, 2, 3, 3)
d_1	(0.2645, 0.2832, 0.4523)	–	–
d_2	(0.1881, 0.2135, 0.3285, 0.2699)	–	–
d_3	(0.197, 0.3151, 0.2172, 0.2707)	–	–
d_1-d_2, d_1-d_3, d_2-d_3	(0.9933, 0.0009, 0.0059)	–	–

Table 12: Essay 19